

Factor VIII and IX assays for post-infusion monitoring in hemophilia patients: Guidelines from the French BIMHO group (GFHT)

Emmanuelle Jeanpierre¹  | Claire Pouplard² | Dominique Lasne³ |
Véronique Le Cam Duchez⁴ | Valérie Eschwege⁵ | Claire Flaujac⁶ | Hubert Galinat⁷ |
Ines Harzallah⁸ | Valérie Proulle⁹ | Motalib Smahi¹⁰ | Frédéric Sobas¹¹ |
Natalia Stepina¹² | Pierre Toulon¹³ | Sophie Voisin¹⁴ | Catherine Ternisien¹⁵ |
Christophe Nougier¹¹ | On behalf of the French Study Group on the Biology of
Hemorrhagic Diseases (the BIMHO group)*

¹CHU Lille, Institut d'Hématologie Transfusion, Institut Pasteur de Lille, UMR1011-EGID, Lille, 59000, France

²Service d'Hématologie Hémostase, Hôpital Trousseau, CHU de Tours, EA 7501, Université François Rabelais, Tours, France

³Laboratoire d'Hématologie Générale, Hôpital Necker AP-HP, Université Paris Sud Saclay, INSERM U1176, Paris, Le Kremlin-Bicêtre, France

⁴CHU de Rouen, UF Hémostase Hématologie Biologique, Rouen, France

⁵Service d'Hématologie Hémostase, CHU de Nancy, France

⁶Laboratoire d'Hématologie, Centre Hospitalier de Versailles, Le Chesnay, France

⁷Laboratoire d'Hématologie, Hôpital La Cavale Blanche, CHU de Brest, France

⁸Service d'Hématologie Hémostase, Groupe Hospitalier Régional de Mulhouse et Sud Alsace, France

⁹Service d'Hématologie Biologique, CHU Bicêtre, AP-HP, Université Paris Sud Saclay, INSERM U1176, Le Kremlin-Bicêtre, France

¹⁰Service d'Hématologie Hémostase, Hôpital Simone Veil, Eaubonne, France

¹¹Service d'Hématologie Hémostase, Hospices Civils de Lyon, Bron, France

¹²Institut Pasteur de Lille, UMRU1011-EGID, Lille, 59000, France

¹³Service d'Hématologie Biologique, Hôpital Pasteur, CHU Nice, France

¹⁴Laboratoire d'Hématologie, CHU de Toulouse, France

¹⁵Service d'Hématologie Hémostase, Hôtel-Dieu, CHU de Nantes, France

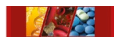
Correspondence

Emmanuelle Jeanpierre, CHU de Lille,
Centre de Biologie Pathologie, Laboratoire
d'Hémostase, bd J Leclercq, 59037 Lille
Cedex, France.
Email: emmanuelle.jeanpierre@chru-lille.fr

Abstract

Replacement therapy with plasma-derived or recombinant FVIII and FIX (pdFVIII/pdFIX or rFVIII/rFIX) concentrates is the standard of treatment in patients with haemophilia A and B, respectively. Measurement of factor VIII (FVIII:C) or factor IX (FIX:C) levels can be done by one-stage clotting assay (OSA) or chromogenic substrate assay (CSA). The French study group on the Biology of Hemorrhagic Diseases (a collaborative group of the GFHT and MHEMO network) presents a literature review and proposals for the monitoring of FVIII:C and FIX:C levels in treated haemophilia A and B patients, respectively. The use of CSA is recommended for the monitoring of patients treated with pdFVIII or rFVIII including extended half-life (EHL) rFVIII.

*See Acknowledgements for the authors present in the BIMHO group.



Except for rFVIII-Fc, great caution is required when measuring FVIII:C levels by OSA in patients substituted by EHL-rFVIII. The OSA is recommended for the monitoring of patients treated with pdFIX or rFIX. Large discordances in the FIX:C levels measured for extended half-life rFIX (EHL-rFIX), depending on the method and reagents used, must lead to great attention when OSA is used for measuring FIX:C levels in patients substituted by EHL-rFIX. Data of most of recent studies, obtained with spiked plasmas, deserve to be confirmed in plasma samples of treated patients.

KEYWORDS

chronometric and chromogenic assays, Factor IX, Factor VIII, haemophilia A, haemophilia B

1 | INTRODUCTION

The variability between the methods used to determine the activities of factor VIII (FVIII:C) and factor IX (FIX:C) has been known for several years, particularly for haemophilia patients substituted with recombinant products.^{1,2} This variability is more important for some extended half-life (EHL) products,³ and the consequences in terms of monitoring are major.

A scientific standardization subcommittee, “FVIII, FIX and rare bleeding disorders,” of the Scientific and Standardization Committee (SSC) of the International Society on Thrombosis and Haemostasis (ISTH)⁴ has issued recommendations to define the quantity of the active substance in the vial and to standardize the potency labelling of FVIII and FIX concentrates.

Currently, the chromogenic assay (CSA) is the international reference for the potency labelling of highly purified FVIII concentrates and has been the reference method of the European Pharmacopoeia since 1994,⁵ whereas FIX concentrates must be titrated only by the one-stage clotting assay (OSA).⁶

From June 2014, the European Medicines Agency (EMA) workshop therefore requires the pharmaceutical companies producing FIX to point out the activated partial prothrombin time (APTT) activator that allows optimal measurement of the product that is best correlated with the activity of the product in vivo (EMA/135928/2014 Human Medicines Development and Evaluation). In laboratories, the most commonly used assay for determining FVIII and FIX activities is the OSA, which is based on APTT; however, the huge variability of the APTT reagents and automater raises difficulties for postinfusion monitoring.⁷⁻⁹

In this study, the French Study Group on the Biology of Hemorrhagic Diseases [the BIMHO group, a collaborative group of the Groupe Français d'étude sur l'Hémostase et la Thrombose (GFHT)] and the French network on inherited bleeding disorders (MHMO) present a literature review, together with proposals for the monitoring of the postinfusion monitoring of patients.

2 | METHODOLOGY

The methodology for the establishment and validation of these proposals was as follows. The different parts of this text were assigned to three working groups, consisting of members of the BIMHO group. Each group made a proposal based on data from the literature after the proposals were reviewed, discussed and modified, and then subjected to critical analysis by all BIMHO members. Finally, the proposals were validated by a vote (39 voters, ie 90% of the members), thereby determining the strength of each proposal, as follows. To make a proposal on an item, at least 50% of the members had to express their agreement (for a strong agreement to be reached, the threshold was set at 7, using a scoring scale between 1 and 10); fewer than 20% gave a score below 4.

The bibliographic search was limited to the English language and was conducted on the PubMed database (from 1989 to January 2020), including analysis of references from the selected articles, with the following keywords: FVIII, FIX, assays, discrepancies, haemophilia and extended half-life.

Only articles containing information concerning the assays used to determine FVIII or FIX activity in patient samples following infusion or in spiked samples were included in the present analysis. For some parts of this review, when data from the literature were insufficient, work presented at congresses (oral or poster papers) during the last 3 years was taken into account.

The proposals were weighted according to the available data. Thus, recommendations or non-recommendations of a particular method were based on the positions of the scientific societies. “Acceptable/unacceptable” methods were based on analysis of the data published in the literature. In the absence of specific calibration, plasma provided and validated by the manufacturers' methods was classified as unacceptable.

When the results were presented only at congresses (oral or poster presentations), the methods were considered as “acceptable but to be confirmed” or “unacceptable but to be confirmed.” In some cases, results in the different publications (summarized in the supplementary tables) were conflicting.



2.1 | Summary of the reagents available for FVIII:C and FIX:C assays

Different reagents for APTT used for the measurement of FVIII and FIX by the OSA, as well as various reagent kits for the CSA, are presented in Table 1.

3 | RESULTS

3.1 | Plasma-derived FVIII and FIX

3.1.1 | Various molecules and their characteristics

The plasma-derived anti-haemophilic factors discussed in this section are as follows: Factane[®] (LFB), Octanate[®] (Octapharma), Betafact[®]

(LFB), Mononine[®] (CSL Behring) and Octafix[®] (Octapharma) which are, respectively, FVIII and FIX.

3.1.2 | Titration of plasma-derived FVIII and FIX by manufacturers

In Europe, the activity of plasma-derived FVIII (pd-FVIII) is measured in international units (IU) and is determined by the CSA against a World Health Organization (WHO) standard according to the recommendations of the European Pharmacopoeia. For the plasma-derived FIX (pd-FIX), activity is determined by the European Pharmacopoeia's OSA compared to the WHO International Standard. The titration of Betafact[®] and Mononine[®] is performed with Pathromtin SL (silica as activator), while Octafix[®] is titrated using an APTT reagent with kaolin as an activator (Pathromtin[®]).

TABLE 1 List of the reagents used in the studies for OSA and CSA

Name	Activator	Phospholipid origin	Current suppliers
APTT-SP	Colloidal silica	Synthetic	Werfen
Cephen	Micronized silica equivalent	Plant	Sysmex
Actin FS	Ellagic acid	Soy	Siemens
Actin FSL	Ellagic acid	Soy and rabbit	Siemens
Actin	Ellagic acid	Rabbit	Siemens
Dapttin	Silica and sulphatide	Blend of highly purified phospholipids	Technoclone
DG Synth	Ellagic acid	Synthetic	Grifols
Pathromtin SL	Silicon dioxide	Plant	Siemens
Pathromtin	Kaolin	Plant	Siemens
Platelin L	Micronized silica	–	Tcoag
Platelin LS	Micronized silica	Porcine and chicken	Tcoag
PTT-A	Micronized silica	Rabbit	Diagnostica Stago
Cephascreen	Polyphenolic component	Rabbit	Diagnostica Stago
CK Prest	Kaolin	Rabbit	Diagnostica Stago
SynthAFax	Ellagic acid	Synthetic	Werfen
SynthASil	Colloidal silica	Synthetic	Werfen
aPTT HS	Micronized silica	Porcine and chicken	Tcoag
aPTT S	Micronized silica	–	Tcoag
Automated aPTT	Silica	Rabbit	Tcoag
Name	Protein origin	Current supplier	
FVIII chromogenic	Bovine	Siemens	
Biophen FVIII	Human	Sysmex	
Coamatic FVIII	Bovine	Chromogenix	
Coatest SP FVIII	Bovine	Chromogenix	
Electrachrome FVIII	Bovine	Werfen	
Technochrom FVIII	Human/bovine	Technoclone	
Trinichrom FVIII	Bovine	Stago	
Biophen FIX	Human	Hyphen Biomed	
Rox FIX	Bovine	Rossix	



3.1.3 | Biological monitoring: available studies

The most commonly used assay for patients treated with pd-FVIII or pd-FIX is still the OSA, and it is recommended to perform calibration using a standard calibrated against the WHO International Standard.

The difference in results between the OSA and CSA in patients' samples treated with pd-FVIII (Factane®) is only approximately 10%, with higher FVIII:C levels using the chromogenic method.⁷

Few studies mentioning pd-FIX assay methods have been published so far.¹⁰ Concerning Mononine®, a pharmacokinetic study¹¹ using the automated APTT Organon (currently marketed as TriniCLOT automated aPTT) for the determination of FIX:C shows measured levels very close to the manufactured label values. Comparable results were published in another study using Pathromtin SL and APTT-SP, but an overestimation of more than 150% of the low value (0.03 IU/mL) was observed with the other reagents (SynthAFax, SynthASil, Actin, Actin FS and Actin FSL).¹² Rox FIX and Biophen FIX chromogenic kits are acceptable for monitoring patients treated with Mononine®.^{12,13}

The method(s) that should be used for the follow-up of patients treated with plasma-derived FVIII or FIX are presented in Table 2.

3.1.4 | Proposal for plasma-derived FVIII

It is recommended to use a CSA, calibrated with a standard calibrated against the WHO International Standard, for the follow-up of patients treated with pd-FVIII. Use of the OSA is acceptable (strong agreement).

3.1.5 | Proposal for plasma-derived FIX

It is recommended to use the OSA for the follow-up of patients treated with of pd-FIX (strong agreement).

3.2 | Conventional recombinant FVIII or FIX (except EHL)

3.2.1 | Various molecules and their characteristics

The characteristics of the different recombinant factors FVIII (rFVIII) and FIX (rFIX) evaluated in this review are presented in Table S1.

3.2.2 | Titration of recombinant factors (except EHL) by manufacturers

In accordance with the recommendations of the EMA, all recombinant FVIII concentrates presented in Table S1 are titrated by the CSA. The titration of Rixubis® and Benefix® vials is performed by the OSA with Dapttin and PTT-A, respectively, as the APTT reagent (Table 1).

3.2.3 | Biological monitoring: available studies

During the monitoring of haemophilic patients treated with recombinant FVIII, many discrepancies were reported between the OSA and CSA. The FVIII:C levels obtained with a chromometric assay during rFVIII treatment can be up to 50% lower than those measured by a chromogenic method.^{5,14} These discrepancies fluctuate according to the composition of the reagent used (activator, phospholipid nature and concentration) in the OSA and the recombinant FVIII measured. Not only the molecular structure of rFVIII, but also the different post-translational modifications from one cell line to another (CHO, BHK, HEK cell lines) can influence the activity measurements. Some studies reported differences between the CSA and OSA for the determination of FIX:C with rFIX.^{12,13,15-17}

3.2.4 | Recombinant FVIII (except EHL)

Octocog alpha—Advate®

Advate® is often used in studies as a recombinant reference FVIII.^{6,18-21} Most studies have reported higher average levels of FVIII by the CSA (+10% to +19%).^{19,22,23} For target values less than 0.05 IU/mL, OSA/CSA ratios of 1:5 have been reported in some studies,^{19,23} while Kitchen et al did not observe differences in low FVIII levels.⁶

Octocog alpha—Kogenate®/Helixate Nexgen®

Differences have been described between the OSA and CSA activities of FVIII:C, particularly when the OSA uses kaolin.²⁴ A multi-centre study conducted from plasma of patients treated with Kogenate®/Helixate Nexgen® showed that lower levels of FVIII:C were obtained by the OSA, with differences of -22% with SynthASil (n = 20) and -27% with Actin FS (n = 15).²⁵ A difference of -17.8% was also observed in the work by Pouplard et al in 2016.²²

TABLE 2 Plasma-derived factors: proposals for the determination of FVIII or FIX after substitution

	pd-FVIII		pd-FIX		
	Factane®	Octanate®	Betafact®	Mononine®	Octafix®
Chromogenic method	Recommended	Recommended	Not evaluated	Acceptable	Not evaluated
Chronometric method	Acceptable	Acceptable	Recommended	Recommended	Recommended

Note: Recommended/non-recommended: position of learned societies; acceptable/unacceptable: data published in the literature; acceptable but to be confirmed/unacceptable but to be confirmed: results that have been only the subject of oral or poster communications; and not evaluated: no study.



Octocog alpha—Kovaltry®

A multi-centre study carried out on Kovaltry®-spiked plasmas validated the OSA and CSA, with average ratios of CSA/OSA between 1:04 and 1:14.²⁶ These results were confirmed in the Leopold I and II efficacy and safety studies.¹⁸

Moroctocog alpha—Refacto AF®

The underestimation of FVIII:C levels in the order of 30%-50% by the OSA, more pronounced with high FVIII:C levels, has been reported by several teams.^{1,14} The activity of FVIII measured by the CSA was well correlated with the antigenic level of FVIII in treated severe haemophilia A patients (ratio 1:03). In 2016, Kitchen et al²⁵ carried out a study on external quality control and showed an overestimation of 27% by the CSA, despite the use of the Refacto AF® standard when Actin was used. No objective differences were found when SynthASil was employed. Several studies have validated the use of a specific calibration plasma for the OSA (standard Refacto AF®) to reduce discrepancies.^{5,14,22,27,28} Nevertheless, the variability of the results according to the aPTT reagent requires great vigilance, and the CSA remains the preferred assay method. The availability of many new drugs makes it impossible to consider the use of a specific calibration plasma for each one. This would increase the number of test configurations and lead to logistical difficulties.

Turoctocog alpha—Novoeight®

An international multi-centre study conducted with deficient plasma spiked with Novoeight® reported the CSA/OSA FVIII:C ratios ranging from 0:68 to 1:30, with an overestimation of low values by the OSA without clinical relevant differences in FVIII levels, depending on the aPTT reagent used, and an overestimation of high values by the chromogenic method.²³ This effect is not specific to the molecule, as the same observation was made with Advate®.^{19,22,23} Different chromogenic kits (Coamatic FVIII, Coatest FVIII, Biophen FVIII, FVIII chromogenic) were also evaluated with two different calibrators (WHO eighth International Standard and Standard Human Plasma (SHP) Siemens) using severe haemophilia A plasma spiked with Novoeight®.²⁹ Regardless of the chromogenic kit, the closest values to the target values were measured with the WHO eighth calibrator.

Simoctocog alpha—Nuwiq®

A study performed by Klukowska et al compared the pharmacokinetic parameters of Nuwiq® obtained with the OSA and CSA. Unfortunately, which reagents were used was not mentioned. The analysis showed that the FVIII:C level was less than 20% higher than the target value with the CSA.³⁰

3.2.5 | Recombinant FIX (except EHL)

Nonacog gamma—Rixubis®

The concentrations of Rixubis® measured by the OSA (Dapttin) or CSA (Rox FIX or Biophen FIX) were closely comparable, according to

Turecek et al (2015).¹⁶ However, the work by Kershaw et al shows that only the Rox FIX chromogenic method can measure FIX levels close to the target values.³¹ Indeed, the use of the OSA with Triniclot aPTT, SynthAFax or Actin overestimated the sample by more than 125% at 0.05 IU/mL, regardless of the aPTT reagent used. Results closest to the target values were obtained with SynthAFax, while Actin and Triniclot aPTT overestimated FIX levels greater than or equal to 0.2 IU/mL. Pathromtin SL was validated in a study presented during the XXXI International Congress of the World Federation of Haemophilia (WFH) in 2014.³² This work reported an overestimation of up to 40% with the other reagents tested (mainly Actin FS, Actin FSL, SynthASil and Cephen). Both chromogenic kits were acceptable according to this study for the monitoring of Rixubis®.

Nonacog alpha—Benefix®

Three major studies evaluated the impact of several aPTT reagents on the determination of FIX using samples spiked with this recombinant FIX.^{12,31,33} The Kershaw study, previously cited for the determination of Rixubis®, validates the Rox FIX chromogenic reagent and the use of SynthAFax for the OSA. However, the Biophen FIX kit underestimated FIX:C levels, with results below 75% of the target value for levels less than or equal to 0.2 IU/mL. The same finding was noted with the samples from treated patients.^{12,31} According to Sommer's study, FIX levels were overestimated by the OSA when ellagic acid was used as an activator, and the levels closest to the target values were obtained with kaolin.³³ Meanwhile, silica provided intermediate results. The variability of high inter-laboratory results of this study should be highlighted, with coefficients of variation (CVs) for the same reagent ranging from 12% for high values (0.8 IU/mL) to 30% for lower values (0.3 IU/mL).

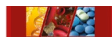
The method(s) that should be used for the follow-up of patients treated with rFVIII or rFIX (except EHL) are presented Table 3.

3.2.6 | Proposal for recombinant FVIII (except EHL)

It is recommended to use the CSA calibrated with a standard connected to the WHO International Standard for the follow-up of patients treated with recombinant FVIII. To monitor a patient treated with Refacto AF®, if the OSA is not calibrated with a specific Refacto AF® standard, it should be compared with the CSA (strong agreement).

3.2.7 | Proposal for recombinant FIX (except EHL)

The OSA calibrated with a standard connected to the WHO International Standard is recommended for the follow-up of patients treated with recombinant FIX. Chromogenic methods are acceptable for the follow-up of patients treated with Rixubis®, and the data are currently insufficient for the follow-up of patients treated with Benefix® (strong agreement).

**TABLE 3** Proposals for the determination of FVIII and FIX after substitution by a recombinant factor

	rFVIII			rFIX	
	rFVIII full length	rFVIII B domain deleted or B domain truncated		rFIX full length	
	Advate [®] , Kogenate [®] , Helixate Nexgen [®] , Kovaltry [®]	Refacto AF [®]	Novoeight [®] Nuwiq [®]	Rixubis [®]	Benefix [®]
Chromogenic assay	Recommended	Recommended	Recommended	Acceptable	Insufficient data
Chronometric assay	Acceptable	Acceptable ^a with a specific Refacto AF standard	Acceptable	Recommended	Recommended

Note: Recommended/not-recommended: position of learned societies; acceptable/not-acceptable: data published in the literature; acceptable but to be confirmed/unacceptable but to be confirmed: results that have only been the subject of oral or poster communications; and not evaluated: absence of a study

^aThe dosage by the chronometric method has been validated using a Refacto AF[®] standard.

3.3 | Recombinant FVIII and FIX with extended half-life

3.3.1 | Various molecules and their characteristics

Strategies used to develop recombinant FVIII or FIX with an extended half-life (EHL) are based on the development of fusion protein (immunoglobulin fragment, recombinant albumin) or chemical modifications of FVIII or FIX with polyethylene glycol (PEG) coupling. A domain B-deleted rFVIII with a single chain has also been developed. Table S2.

3.3.2 | Titration of recombinant EHL factors by manufacturers

In accordance with the recommendations of the EMA, rFVIII-EHLs are titrated by CSA. FIX-EHL vials are titrated by the OSA using Actin or Pathromtin SL for Alprolix[®] or Idelvion[®] vials, respectively. Refixia[®] is titrated by an OSA with SynthAFax and a calibrator of human origin.

3.4 | Biological monitoring: available studies

3.4.1 | Recombinant FVIII-EHL

Efmoroctocog—Elocta[®]/Eloctate[®]

During international clinical studies, assays were centralized and performed by both OSA (Actin FSL) and CSA^{34–36} (Table S3). Higher FVIII activities were obtained through the chromogenic method with a bias of 21 and 32% for Advate[®] and Elocta[®], respectively.³⁵ This effect was more pronounced for concentrations above 0.5 IU/mL.³⁵ Comparable results have also been reported when CK Prest was used, with an average bias of approximately 10%, not exceeding 20% for rFVIII-Fc concentrations above 1 IU/mL.³⁷

A multi-centre *in vitro* study (30 laboratories) was performed on spiked deficient FVIII plasma with increasing levels of Elocta[®] or Advate[®]. The assays were carried out with reagents specific to each laboratory for the OSA (ellagic acid, silica, kaolin and polyphenol) and four different kits for the CSA (Biophen FVIII, FVIII chromogenic, Technochrom FVIII and Coatest SP FVIII). The two recombinant FVIII (EHL or non-EHL) behaved in the same way whether they were measured by the OSA or CSA methods.²¹ The APTT reagent used did not seem to have an impact on the results, but details of the results obtained with each reagent were not mentioned in the article. These *in vitro* results are in agreement with the studies conducted on patients' plasma treated with rFVIII-Fc, showing an excellent correlation between the two methods for FVIII assays, regardless of the reagents used. Underestimation of the results by the OSA compared with CSA is clinically acceptable. The results obtained *in vitro* and *ex vivo* with rFVIII-Fc show that the relative error is less than 25%, regardless of the reagent/automated pair used without using a molecule-specific standard for calibration.

3.5 | Which method(s) should be used for follow-up of patients treated with Elocta[®]?

The chromogenic method is recommended. The OSA is acceptable (strong agreement).

3.5.1 | Bax-855—Adynovi[®] and Adynovate[®]

An international multi-centre study (35 laboratories) conducted on deficient plasmas spiked with Adynovi[®] and Advate[®] was published by Turecek et al in 2016.¹⁹ The results obtained by the OSA were very close to the target values, regardless of the APTT activator used (silica, kaolin, ellagic acid or polyphenol). The CSA was performed only by 11 laboratories, and the results obtained were, on average, 14% higher than the target value (Table S4). An independent Swiss study (eight laboratories) compared the CSA with



the OSA using different calibration curves (conventional standard plasma or a specific standard prepared with Adynovi®) and different APTT activators (silica, kaolin or ellagic acid).³⁸ When calibrations were performed with a conventional standard plasma, the authors obtained a systematic overestimation of FVIII levels using either the OSA (+27 to 113%) or the CSA (+11 to 83%). Only the use of a specific standard can allow measuring FVIII levels close to the expected values (Table S4).

3.6 | Which method(s) should be used for the follow-up of patients treated with Adynovi®?

The chromogenic method is recommended. The published data are insufficient and/or contradictory to make proposals regarding the OSA (strong agreement).

3.6.1 | BAY 94-9027—Jivi®

According to the Gu et al study, the PEGylated rFVIII BAY 94-9027 could disrupt coagulation tests performed with some silica-based reagents.³⁹ Indeed, on samples spiked with this rFVIII-PEG, FVIII:C activities measured with reagents based on silica (APTT-SP, PTT-A), polyphenol (Cephascreen) or ellagic acid (SynthAFax, Actin) were only close to the target values with reagents containing ellagic acid. Values close to the target values were also obtained by the CSA.

In 2018, Church et al validated the use of SynthASil (results between 118% and 103% of target values), Pathromtin SL (results between 129% and 111% of target values) and Actin FSL (results between 107% and 89% of target values).⁴⁰ In this study, the type of activator alone did not appear to be responsible for the differences in FVIII:C levels observed for patients substituted with Jivi®. Indeed, while Actin FS induced an overestimation of FVIII:C levels by more than 40% compared with the target values, Actin FSL allowed rates to be closer to the expected values. CK Prest and Pathromtin SL were not validated for rates below 0.5 IU/mL, because they overestimated the results. The chromogenic kits (FVIII chromogenic, Coamatic FVIII and Electrachrome FVIII) are usable according to this study⁴⁰ (Table S5).

3.7 | Which method(s) should be used for the follow-up of patients treated with Jivi®?

The CSA is recommended. The published data are insufficient and/or contradictory to make proposals regarding the OSA (strong agreement).

3.7.1 | Turoctocog alpha pegol-N8-GP—Esperoct®

The study by Pickering et al carried out on plasma spiked with Esperoct® shows good performance of four chromogenic assays:

Coamatic FVIII, Coatest SP FVIII, Biophen FVIII and FVIII chromogenic.²⁹ These results were not confirmed by the work carried out in two European centres with the same four reagents, which highlighted results higher than the target value of more than 10%-40%. The OSA carried out with Actin, Actin FS and Pathromtin SL made it possible to obtain FVIII:C levels very close to the target values. However, an underestimation of more than 25% was observed with Actin FSL, SynthASil and SynthAFax for target values greater than or equal to 0.2 IU/mL. This effect is more pronounced with APTT-SP⁴¹ (Table S6).

3.8 | Which method(s) should be used for the follow-up of patients treated with Esperoct®?

Published data on the CSA validate the use of some kits only; the published data on the OSA are insufficient to submit proposals (strong agreement).

3.8.1 | Lonoctocog alpha—Afstyla®

The manufacturer of this drug in the summary of product characteristics (SPC) recommends the CSA for monitoring treated patients or applying a correction factor of 2 if the FVIII:C level is measured by the OSA. A recent multi-centre study carried out by 23 laboratories showed that on samples spiked with Afstyla®, an underestimation of the FVIII:C level by OSA was observed, regardless of the measurement range (between 0.04 and 1 IU/mL) and the type of activator reagents used (kaolin, silica, ellagic acid and silicon dioxide).²⁰ However, according to Bowyer et al, the multiplication factor of 2 leads to a very significant relative overestimation (>100%) of the low rates, close to 0.02 IU/mL, and may induce, depending on the measurement area, overestimation of the levels with APTT-SP, SynthAFax or Actin FS and underestimation with SynthASil.⁴²

3.9 | Which method(s) should be used for the follow-up of patients treated with Afstyla®?

The CSA is recommended. Concerning the OSA, an underestimation of approximately 50% is reported. If the OSA is used to monitor a patient, it must have been compared with the CSA (strong agreement).

The method(s) that should be used for the follow-up of patients treated with rFVIII-EHL are presented in Table 4.

3.9.1 | Proposal for recombinant FVIII-EHL

The CSA is recommended for all rFVIII half-life. The OSA is currently only acceptable for Elocta®. Great care must be taken when using the OSA for the determination of other rFVIII-EHL (strong agreement).

**TABLE 4** Proposals for the determination of FVIII:C after substitution by an rFVIII-EHL

	rFVIII-Fc	rFVIII B deleted or B truncated PEG			rFVIII Single chainroncated
	Elocta [®]	Adynovi [®]	BAY 94-9027	N8-GP	Afstyla ^{®a}
Chromogenic assays	Recommended	Recommended	Recommended	Recommended only with some kits	Recommended
Chronometric assays	Acceptable	Discrepant data	Unacceptable: reagents PTT-A et APTT-SP Discrepant data with other APTT reagents	Discrepant data	Acceptable only after comparison with the recommended method

Note: Recommended/non-recommended: position of scientific societies; acceptable/unacceptable: data published in the literature; acceptable but to be confirmed/unacceptable but to be confirmed: results only accepted as oral or poster communications; and not evaluated: absence of a study. Discrepancies in data: different conclusions according to the studies.

^aA correction factor of two is proposed by the RCP, but it is not universal.

3.9.2 | Recombinant FIX-EHL

Eftrenonacog—Alprolix[®]

A multi-centre study conducted on FIX-deficient plasmas spiked with Alprolix[®] at 0.05–0.20 and 0.80 IU/mL and distributed in 30 laboratories highlighted an underestimation of 50% when FIX:C activities were measured with kaolin compared with expected values obtained with Actin.³³ Pathromtin SL, SynthASil and automated aPTT were acceptable for the determination of FIX:C, with differences of less than 15% from the target values. In contrast, differences observed with PTT-A vary widely from one centre to another, and in some cases can reach more than 30%.⁴³ These results were confirmed in a British external quality assessment programme [UK National External Quality Assessment Service (NEQAS) exercise 2016] presented at the XXXIII International Congress of the WFH by Kitchen et al and in the Dericquebourg et al study presented in 2018 at the 11th Annual European Association for Haemophilia and Allied Disorders (EAHAD) Congress.⁴⁴ The last two studies validated the two chromogenic kits (Biophen FIX and Rox FIX) to measure Alprolix[®] activity.

A European study presented by Willemze et al at the same Congress and conducted in 59 centres⁴⁵ revealed that, with CK Prest (used by 23 centres and only on Diagnostica Stago[®] instruments), an underestimation of the FIX:C was observed for target values between 0.20 and 0.80 IU/mL (20.5% and 27.5%, respectively). A wide variability in the differences was reported from one centre to another (extreme values: 12.5%–43%). The Actin FS used by 14 centres on the Sysmex[®] instrument overestimated the levels by 18% and 25%, respectively. The SynthASil used by 10 centres on the Werfen[®] instrument provided levels very close to the target values. Performances of some reagents seemed to be dependent upon the reagent/instrument pair. Thus, the Pathromtin SL used on a Siemens[®] instrument overestimated the results by more than 25%, but this effect was not observed if this reagent was used on a Sysmex[®] instrument (four users) or a Diagnostica Stago[®] instrument (one user).

Only one study presented in the 11th Annual EAHAD Congress by Bowyer et al⁴⁶ was performed on samples from five patients

treated with Alprolix[®]. Out of 22 samples, taking as a reference value the rates measured with Actin, the authors showed an underestimation of approximately 20% with Actin FS, Actin FSL, SynthASil and SynthAFax. Differences of more than 20% were also observed with Pathromtin SL and APTT-SP (25% and 28%, respectively). With the CSA, the Biophen FIX reagent was underestimated by 32.2%, while the levels measured with the Rox FIX reagent were very close to the target values.

3.10 | Which method(s) should be used for the follow-up of patients treated with Alprolix[®]?

Several APTT reagents are validated for the determination of FIX:C for the biological monitoring of Alprolix[®] treatments (Table S7). Regarding the CSA, the Rox FIX kit seems to be better suited for patients substituted by this product.

CK Prest, PTT-A and Pathromtin SL induced an underestimation of FIX levels, but with differences varying from one laboratory to another. Therefore, it is necessary, in case of the use of one of these three reagents, to have previously compared the measured FIX rates with a validated method.

3.10.1 | Albutrepenonacog—Idelvion[®]

At the XXXII International Congress of the WFH in 2016, St Ledger et al⁴⁷ presented the results of two *ex vivo* studies carried out on FIX-deficient plasma spiked with Idelvion[®]. These samples were evaluated by one central laboratory with Pathromtin SL and by participating laboratories using their own APTT reagents: Pathromtin SL, aPTT HS, Platelin LS, APTT-SP and CK Prest. The majority of APTT reagents provided results no more than 20% different from those obtained in the central laboratory. However, an underestimation of approximately 50% was observed, regardless of the measurement domain, with Actin FS and CK Prest.

Similar results were presented at the 62nd ISTH SSC meeting in 2016 by Buyue et al⁴⁸. Those results are also comparable with the



results of the UK NEQAS (exercise 2016), showing an underestimation of more than 50% with the CK Prest and of approximately 30% with Actin FSL, SynthASil and PTT-A. Unsurprisingly, Pathromtin SL, used by the central laboratory of CSL Behring, provided the expected rates. During this exercise, the two chromogenic kits (Rox FIX and Biophen FIX) evaluated significantly overestimated the FIX levels (+93% and +65%, respectively).

A study by the BIMHO group evaluated plasmas spiked with Idelvion® and titrated with Pathromtin SL (0.05-0.3 and 0.8 IU/mL).⁴⁹ A systematic underestimation of 50% was obtained with the CK Prest for samples at 0.3 and 0.8 IU/mL. Greater variability of results was observed for the sample titrated at 0.05 IU/mL (measured concentrations between 0.02 and 0.06 IU/mL). Three studies presented at the 64th ISTH SSC meeting in 2018 showed an overestimation of FIX:C levels by more than 35% by chromogenic methods (Biophen FIX and Rox FIX).^{43,50,51}

A single study performed on 18 samples from three patients treated with Idelvion® was presented at the 11th Annual EAHAD Congress by Bowyer et al.⁴⁶ Differences of less than 10% were observed with Actin, APTT-SP and SynthASil. Meanwhile, an underestimation of more than 30% was observed with Actin FS and Actin FSL. An overestimation of nearly 50% was observed with SynthAFax and with the two chromogenic kits (Biophen FIX and Rox FIX).

Most recently, an underestimation close to 50% of FIX:C levels with Actin FS and CK Prest has been confirmed in a multi-centre study by Horn et al.⁵² The results obtained with SynthASil and Pathromtin SL were very close to the expected values, and the differences obtained with APTT-SP, STA-PTT, Actin and Cephascreen were less than 25%.

3.11 | Which method(s) should be used for the follow-up of patients treated with Idelvion®?

Given the current state of knowledge, the two chromogenic kits (Biophen FIX and Rox FIX) are not acceptable for the determination of Idelvion® activity.

Only the OSA can be used, and a large number of APTT reagents have been validated to date: Actin, Pathromtin SL, APTT-SP, PTT-A, SynthASil and Platelin LS. When these reagents are not available in the laboratory, it is important to take into account underestimation of the FIX:C levels measured with CK Prest and Actin FS. In the French study conducted with the CK Prest, the variability of the results, observed especially for the low concentrations, does not validate a systematic correction factor (Table S8).

3.11.1 | Nonacog beta pegol—Refixia®

In a study published in 2016 by Bowyer et al and conducted in two centres on deficient plasma spiked with Refixia® (target values of 0.03-0.90 IU/mL), the authors validated the CSA (Rox FIX and Biophen FIX) for all concentrations of Refixia®, with a recovery

percentage of between 74% and 126%.¹² One-stage assays allowed results close to the target values only with SynthAFax and DG-APTT Synth. With SynthASil, Actin, Actin FS and Actin FSL, the results were underestimated by 50%-70%.

Conversely, Pathromtin SL and APTT-SP, whose activator is silica, induced an overestimation of the FIX:C by more than 500%, regardless of the target value. This observation was explained by Rosén et al's work,⁵³ showing that Refixia® in the presence of IX-deficient plasma was prematurely converted to FIXa during the activation phase of the OSA when the reagents contained silica. The PEG group made it possible to co-locate Refixia® with its activators (FXIa and Prekallikrein) on the surface of the silica, thus facilitating its early conversion to FIXa during the activation step. For other reagents with other activators (CK Prest, Actin FS), underestimation could be explained by slower activation of Refixia® related to steric hindrance.⁵⁴

A study published by Young et al and performed on plasmas from treated patients confirmed the possible use of the SynthAFax and the CSA (Biophen FIX).⁵⁵

On Refixia®-spiked plasma with concentrations between 0.05 and 2.50 IU/mL, Tiefenbacher et al validated the use of Cephascreen and the two chromogenic kits.⁵⁶ Only these three reagents are recommended by Novo Nordisk in the Refixia® US Food and Drug Administration (FDA) dossier.

3.12 | Which method(s) should be used for the follow-up of patients treated with Refixia®?

There are currently few validated APTT reagents for the determination of Refixia® (Table S9). Chromogenic methods also are validated.

3.13 | Which method(s) should be used for the follow-up of patients treated with rFIX-EHL?

A summary of the results obtained with rFIX-EHL is presented in Table 5.

3.13.1 | Proposal for recombinant FIX-EHL

Studies published on this subject highlight the extremely high heterogeneity in the behaviour of the different APTT reagents with rFIX-EHL. The OSA must be used with the highest caution. Moreover, the majority of published results were obtained on spiked plasma, and discrepant results may be observed on treated patients' plasma. Chromogenic methods are not validated for all these rFIX-EHLs (strong agreement).

4 | DISCUSSION

The emergence of new anti-haemophilic factors with an extended half-life demonstrates significant progress in patient management.

**TABLE 5** Proposals for the determination of FIX:C after substitution by rFIX-EHL

	Alprolix®	Idelvion®	Refixia®
Chronometric method			
Actin	Acceptable	Acceptable	Unacceptable
Actin FS	Acceptable but to be confirmed	Unacceptable	Unacceptable
Actin FSL	Acceptable	Unacceptable	Unacceptable
SynthAFax	Acceptable	Unacceptable	Acceptable
DG Synth	Unacceptable	Not evaluated	Acceptable
PTT-A	Unacceptable but to be confirmed	Acceptable	Not evaluated
APTT-SP	Acceptable but to be confirmed	Acceptable	Unacceptable
SynthASil	Acceptable	Acceptable	Unacceptable
aPTT HS	Unacceptable	Unacceptable – to be confirmed	Not evaluated
Platelin L	Acceptable	Acceptable	Not evaluated
Automated APTT	Acceptable	Not evaluated	Not evaluated
Pathromtin SL	Unacceptable but to be confirmed	Acceptable	Unacceptable
Cephascreeen	Acceptable but to be confirmed	Acceptable	Acceptable
CK Prest	Unacceptable	Unacceptable	Not evaluated
Chromogenic method			
Biophen FIX	Unacceptable but to be confirmed ^a	Unacceptable	Acceptable
Rox FIX	Acceptable	Unacceptable	Acceptable

Note: Recommended/non-recommended: the position of learned societies; acceptable/unacceptable: data published in the literature; acceptable but to be confirmed/unacceptable but to be confirmed: results that have only been the subject of oral or poster communications; and not evaluated: the absence of study.

^aBowyer's results and preliminary results obtained by our group on samples of treated patients do not validate this reagent.

Nevertheless, it represents a real challenge for laboratories involved in the monitoring of substituted patients.

The modification of coagulation factor concentrates, and more precisely the modifications aimed at reducing their clearance from the circulation, has made it difficult to measure FVIII/FIX activity with different results, depending on the assay method (CSA or OSA) and on the reagents used for the OSA.

The differences observed between potencies values obtained using the OSA or CSA or between APTT reagents can reach more than 50%.^{27,49,52} These significant discrepancies could be unacceptable, depending on the factor level and the indication of the dosage. Indeed, for a relative bias of 20% between the two methods we will obtain a difference of approximately 0.20 IU/mL (ie 1 IU/mL $\pm$ 0.20 IU/mL) when a high level of factor is measured, whereas, for the same bias, if the factor level measured is less than 0.1, the difference will reach 0.02 IU/mL.

As noted in the UK guidelines from Gray et al, if one assay consistently gives results 20% below the target based on potency, while another is 20% above, then both may be suitable despite the 40% difference between them.⁵⁷ Currently, there is no definition of an "acceptable difference" or "acceptable relative bias" in the literature. Most teams tolerated a difference when mean results were within 20%-30% of the expected result.^{58,59} Definition of the maximum acceptable bias between the results of FVIII or FIX assays is essential not only for the diagnosis of haemophilia to define the disease severity, but also for the efficient monitoring of treated haemophilic

patients. Indeed, the accuracy of FVIII or FIX activity assays allows an accurate dose adjustment of the FVIII and FIX treatments, and is essential to avoid overtreatment or subtreatment. In this context, there is no clearly defined threshold in the literature, but classically, the preoperative target FVIII level or FIX level is 0.80-1.10 IU/mL for surgery with a major risk of bleeding and 0.50-0.80 IU/mL for surgery with a moderate risk of bleeding.⁶⁰ Knowing the bias for samples containing medium levels and high levels of FVIII or FIX of approximately 0.50 and 1 IU/mL, respectively, is therefore essential.

FVIII and FIX replacement products are assigned a potency label to quantify the amount of active product and guide clinicians on treatment dosing.^{4,15} Underestimating needs can lead to bleeding complications, while overestimating can lead to thrombotic complications and also significant unnecessary costs. Discrepancies in measurement are therefore not simply an analytical concept, but can have important medico-economic consequences. For some rFVIII-EHL and rFVIII products, one-stage reagents are entirely unsuitable for monitoring patients, and in order to minimize differences, the BIMHO group recommends using the CSA, which is the reference method for the determination of FVIII:C in substituted haemophilic A patients.⁶⁰⁻⁶²

Regarding FIX products, some one-stage assays are not acceptable for measurements of postinfusion plasma samples for modified rFIX,^{46,49} and FIX chromogenic assays that have been developed more recently do not provide acceptable results with all EHL-FIX.^{46,48,60} Indeed, the different chromogenic assays are not equivalent. Consequently, it is very important that each clinical laboratory



knows the limitations of the different assay methods that can be used to monitor infusion of EHL-FIX.

As an alternative, calibration curves for the OSA could be performed with standard-specific products, as originally proposed for Refacto-AF[®].^{14,22,28} Unfortunately, these specific products are not always available and are not supplied by the EHL manufacturers. In addition, the laboratory must be fully aware of the treatment received by the patient in order to perform the assays on the appropriate calibration curves. Such an approach can only be envisaged when specific standards are supplied by the manufacturer in highly specialized centres.⁶²

The correction factor that may be proposed for some products is not recommended, as variability not only concerns the activator of the APTT reagents, phospholipids, but also automates and targets value.

The laboratory medical staff are required to be aware of these observations when monitoring substituted haemophiliacs. The particularities of these new molecules with regard to the methods of determination FVIII:C and FIX:C are not always well characterized before they are commercialized, despite international recommendations.⁴ The impact of the activator and phospholipids on the determination of FIX has been reported previously.⁴⁹ Therefore, when changing reagents, the reagent comparison step should ideally include samples of substituted patients. The development of external quality control programmes, including testing of postinfusion samples encompassing all factor replacement products, is desirable.⁵⁶ In emergency situations, if a recommended method is not available 24 hours a day, laboratories may be required to apply a method considered "unacceptable" in our proposals, but only if they have compared it locally in advance with the recommended or an alternative acceptable method. In these conditions, the results obtained with the two methods will be different, and the question arises of transmitting an often-underestimated result to clinicians in the written results report. All the contributors to this text agree that biological results must not be modified by any correction factor before being transmitted. An interpretation of the results describing the observed differences from an acceptable or recommended method must be included in the biological report. Under these conditions, it is therefore essential for the laboratory medical staff to be informed of the anti-haemophilic treatment administered to the patient in order to interpret the result accurately.

5 | CONCLUSION

The aim of these proposals is to help the laboratory medical staff to more clearly understand the performance and limitations of the different assay methods that can be used to monitor infusion of clotting factor concentrates in patients with haemophilia A or B.

The available literature data and biological assays are likely to evolve, and these proposals may be challenged by new publications and the development of new technologies. A number of studies cited in this document have been performed on spiked plasma samples, and the results must be confirmed in samples from treated patients.

ACKNOWLEDGEMENTS

The authors gratefully thank all the members of the BIMHO group: Martine Alhenc Gelas (CHU HEGP, Paris), Christine Biron (CHU Montpellier), Florence Blanc-Jouvan (CH Annecy), Evelyne Bourgerette (CH Nevers), Benedicte Bulabois (CHU Grenoble), Emilie Comio (CH Chambéry), Magali Donnard (CHU Limoges), Jérôme Duchemin (CHU Cochin, Paris), Anne-Camille Faure (CHU St-Etienne), François Grand (CHU Poitiers), Lelia Grunebaum (CHU Strasbourg), Maryse Guicheteau (CHU Poitiers), Felipe Guerrero (CHU Toulouse), Nathalie Hezard (CHU Marseille), Marie-Françoise Hurtaud (CHU Robert-Debré Paris), Fabienne Nedelec Gac (CHU Rennes), Emmanuel De Maistre (CHU Dijon), Raphael Marlu (CHU Grenoble), Guillaume Mourey (EFS Bourgogne Franche Comté), Perrine Munier (CH Valence), Fabienne Pineau-Vincent (CH Le Mans), Didier Raffenot (CH Chambéry), Yohan Repesse (CHU Caen), Anne Ryman (CHU Bordeaux), Laurent Sattler (CHU Strasbourg), Pauline Sauguet (CH Ajaccio), Anne-Francoise Serre-Sapin (CHU Clermont-Ferrand), Alain Stepanian (CHU Lariboisière Paris), Jean Szymezak (CHU Reims), Marie Toussaint-Hacquart (CHU Nancy), Marie Tuffigo (CHU Angers) and Annelise Voyer (CHU Amiens).

ORCID

Emmanuelle Jeanpierre  <https://orcid.org/0000-0002-7900-0518>

REFERENCES

1. Caron C, Dautzenberg M-D, Delahousse B, et al. A blinded in vitro study with Refacto mock plasma samples: similar FVIII results between the chromogenic assay and a one-stage assay when using a higher cephalin dilution. *Haemophilia*. 2002;8:639-643.
2. Dawson NJ, Kemball-Cook G, Barrowcliffe TW. Assay discrepancies with highly purified factor VIII concentrates. *Haemostasis*. 1989;19:131-137.
3. Van den Bossche D, Peerlinck K, Jacquemin M. New challenges and best practices for the laboratory monitoring of factor VIII and factor IX replacement. *Int J Lab Hematol*. 2018;40:21-29.
4. Hubbard AR, Dodd J, Lee T, et al. Recommendations on the potency labelling of factor VIII and factor IX concentrates. *J Thromb Haemost*. 2013;11:988-989.
5. Barrowcliffe TW, Raut S, Sands D, Hubbard AR. Coagulation and chromogenic assays of factor VIII activity: general aspects, standardization, and recommendations. *Semin Thromb Hemost*. 2002;28:247-256.
6. Kitchen S, Kershaw G, Tiefenbacher S. Recombinant to modified factor VIII and factor IX – chromogenic and one-stage assays issues. *Haemophilia*. 2016;22:72-77.
7. Adcock DM, Strandberg K, Shima M, Marlar RA. Advantages, disadvantages and optimization of one-stage and chromogenic factor activity assays in haemophilia A and B. *Int J Lab Hematol*. 2018;40:621-629.
8. Martinuzzo M, Barrera L, Rodriguez M, D'Adamo MA, López MS, Otaso JC. Do PT and APTT sensitivities to factors' deficiencies calculated by the H47-A2 2008 CLSI guideline reflect the deficiencies found in plasmas from patients? *Int J Lab Hematol*. 2015;37:853-860.
9. Toulon P, Eloit Y, Smahi M, et al. In vitro sensitivity of different activated partial thromboplastin time reagents to mild clotting factor deficiencies. *Int J Lab Hematol*. 2016;38:389-396.
10. Aznar JA, Cabrera N, Matysiak M, et al. Pharmacokinetic study of a high-purity factor IX concentrate (Factor IX Grifols) with a 6-month



- follow up in previously treated patients with severe haemophilia B. *Haemophilia*. 2009;15:1243-1248.
11. Berntorp E, Björkman S, Carlsson M, Lethagen S, Nilsson IM. Biochemical and in vivo properties of high purity factor IX concentrates. *Thromb Haemost*. 1993;70:768-773.
 12. Bowyer AE, Hillarp A, Ezban M, Persson P, Kitchen S. Measuring factor IX activity of nonacog beta pegol with commercially available one-stage clotting and chromogenic assay kits: a two-center study. *J Thromb Haemost*. 2016;14:1428-1435.
 13. Wilmot HV, Hogwood J, Gray E. Recombinant factor IX: discrepancies between one-stage clotting and chromogenic assays. *Haemophilia*. 2014;20:891-897.
 14. Mikaelsson M, Oswaldsson U. Assaying the circulating factor VIII activity in haemophilia A patients treated with recombinant factor VIII products. *Semin Thromb Hemost*. 2002;28:257-264.
 15. Dodt J, Hubbard AR, Wicks SJ, et al. Potency determination of factor VIII and factor IX for new product labelling and postinfusion testing: challenges for caregivers and regulators. *Haemophilia*. 2015;21:543-549.
 16. Turecek PL, Abbühl B, Tangada SD, et al. Nonacog gamma, a novel recombinant factor IX with low factor IXa content for treatment and prophylaxis of bleeding episodes. *Expert Rev Clin Pharmacol*. 2015;8:163-177.
 17. Kitchen S, Gray E, Mertens K. Monitoring of modified factor VIII and IX products. *Haemophilia*. 2014;20:36-42.
 18. Kitchen S, Katterle Y, Beckmann H, Maas EM. Chromogenic assay for BAY 81-8973 potency assignment has no impact on clinical outcome or monitoring in patient samples. *J Thromb Haemost*. 2016;14:1192-1199.
 19. Turecek PL, Romeder-Finger S, Apostol C, et al. A world-wide survey and field study in clinical haemostasis laboratories to evaluate FVIII: C activity assay variability of ADYNOVATE and OBIZUR in comparison with ADVATE. *Haemophilia*. 2016;22:957-965.
 20. St. Ledger K, Feussner A, Kalina U, et al. International comparative field study evaluating the assay performance of AFSTYLA in plasma samples at clinical hemostasis laboratories. *J Thromb Haemost*. 2018;16:555-564.
 21. Sommer JM, Moore N, McGuffie-Valentine B, et al. Comparative field study evaluating the activity of recombinant factor VIII Fc fusion protein in plasma samples at clinical haemostasis laboratories. *Haemophilia*. 2014;20:294-300.
 22. Pouplard C, Ternisien C, Desconclois C, Lasne D, Aillaud M-F, Caron C. Discrepancies between one stage assay and chromogenic substrate assay in patients treated with recombinant or plasma-derived FVIII and usefulness of a specific standard in ReFacto AF®-treated patients. *Haemophilia*. 2016;22:e101-e103.
 23. Viuff D, Barrowcliffe T, Saugstrup T, Ezban M, Lillicrap D. International comparative field study of N8 evaluating factor VIII assay performance. *Haemophilia*. 2011;17:695-702.
 24. Lusher JM, Hillman-Wiseman C, Hurst D. In vivo recovery with products of very high purity – assay discrepancies. *Haemophilia*. 1998;4:641-645.
 25. Kitchen S, Jennings I, Makris M, Kitchen DP, Woods TA, Walker ID. Factor VIII assay variability in postinfusion samples containing full length and B-domain deleted FVIII. *Haemophilia*. 2016;22:806-812.
 26. Kitchen S, Beckmann H, Katterle Y, Bruns S, Tseneklidou-Stoeter D, Maas EM. BAY 81-8973, a full-length recombinant factor VIII: results from an international comparative laboratory field study. *Haemophilia*. 2016;22:e192-e199.
 27. Pouplard C, Caron C, Aillaud MF, et al. The use of the new ReFacto AF Laboratory Standard allows reliable measurement of FVIII: C levels in ReFacto AF mock plasma samples by a one-stage clotting assay. *Haemophilia*. 2011;17:e958-e962.
 28. Cauchie M, Toelen J, Peerlinck K, Jacquemin M. Practical and cost-effective measurement of B-domain deleted and full-length recombinant FVIII in the routine haemostasis laboratory. *Haemophilia*. 2013;19:e133-e138.
 29. Pickering W, Hansen M, Kjalke M, Ezban M. Factor VIII chromogenic assays can be used for potency labeling and postadministration monitoring of N8-GP. *J Thromb Haemost*. 2016;14:1579-1587.
 30. Klukowska A, Szczepański T, Vdovin V, Knaub S, Jansen M, Liesner R. Novel, human cell line-derived recombinant factor VIII (Human-cl rhFVIII, Nuwiq®) in children with severe haemophilia A: efficacy, safety and pharmacokinetics. *Haemophilia*. 2016;22:232-239.
 31. Kershaw GW, Dissanayake K, Chen VM, Khoo T-L. Evaluation of chromogenic factor IX assays by automated protocols. *Haemophilia*. 2018;24:492-501.
 32. Gritsch H, Romeder-Finger S, Scheiflinger F, Turecek PL. Potency assignment and measurement of recombinant FIX activity in human plasma impact of reagents on the 1-stage clotting assay. *Haemophilia*. 2014;Suppl 3:37.
 33. Sommer J, Buyue Y, Bardan S, et al. Comparative field study: impact of laboratory assay variability on the assessment of recombinant factor IX Fc fusion protein (rFIXFc) activity. *Thromb Haemost*. 2014;112:932-940.
 34. Mahlangu J, Powell JS, Ragni MV, et al. Phase 3 study of recombinant factor VIII Fc fusion protein in severe haemophilia A. *Blood*. 2014;123:317-325.
 35. Powell JS, Josephson NC, Quon D, et al. Safety and prolonged activity of recombinant factor VIII Fc fusion protein in haemophilia A patients. *Blood*. 2012;119:3031-3037.
 36. Young G, Mahlangu J, Kulkarni R, et al. Recombinant factor VIII Fc fusion protein for the prevention and treatment of bleeding in children with severe haemophilia A. *J Thromb Haemost*. 2015;13:967-977.
 37. Pouplard C, Sattler L, Ryman A, et al. Multicentre pharmacokinetic evaluation of rFVIII-Fc (efmoroctocog alpha) in a real life and comparison with non-extended half-life FVIII concentrates. *Haemophilia*. 2020;26(2):282-289.
 38. Bulla O, Poncet A, Alberio L, et al. Impact of a product-specific reference standard for the measurement of a PEGylated rFVIII activity: the Swiss Multicentre Field Study. *Haemophilia*. 2017;23:e335-e339.
 39. Gu J-M, Ramsey P, Evans V, et al. Evaluation of the activated partial thromboplastin time assay for clinical monitoring of PEGylated recombinant factor VIII (BAY 94-9027) for haemophilia A. *Haemophilia*. 2014;20:593-600.
 40. Church N, Leong L, Katterle Y, et al. Factor VIII activity of BAY 94-9027 is accurately measured with most commonly used assays: results from an international laboratory study. *Haemophilia*. 2018;24:823-832.
 41. Hillarp A, Bowyer A, Ezban M, Persson P, Kitchen S. Measuring FVIII activity of glycopegylated recombinant factor VIII, N8-GP, with commercially available one-stage clotting and chromogenic assay kits: a two-centre study. *Haemophilia*. 2017;23:458-465.
 42. Bowyer A, Key N, Dalton D, Kitchen S, Makris M. The coagulation laboratory monitoring of Afstyl single-chain FVIII concentrate. *Haemophilia*. 2017;23:e469-e470.
 43. Persson A, Mahmoud Hourani Soutari N, Norberg E-M, Antovic J. One stage and chromogenic assays discrepancy in the estimation of factor IX (FIX) recovery. What is it that a single haemophilia center can and cannot do? *Res Pract Thromb Haemost*. 2018;2:120.
 44. Dericquebourg A, Jourdy Y, Negrier C, Nougier C. Comparison of three assays for measuring factor IX. *Haemophilia*. 2018;24:51.
 45. Willemze A, Sadeghi-Khomami A, Sörskog L, Wikén M, Lethagen S. Assessment of clotting activity of recombinant fix FC fusion protein in European haemophilia treatment centers. *Haemophilia*. 2018;24:55.
 46. Bowyer A, Sampson B, Shepherd F, Jones R, Kitchen S, Makris M. The FIX C assessment of extended half-life recombinant factor IX products in clinical practice. *Haemophilia*. 2018;24:56.



47. St Ledger K, Fuessner A, Kalina U, et al. Performance of a recombinant fusion protein linking coagulation factor IX with albumin (rIX-FP) in the one-stage assay. *Haemophilia*. 2016;22:59.
48. Buyue Y, Rabinovich D, Bardan S, et al. Product-specific calibration standards do not correct one-stage clotting assay discrepancies for modified recombinant FIX molecules with reagents that significantly over- or under-recover labeled potency. Annual Meeting of the Scientific Standardization Committee of the International Society on Thrombosis and Haemostasis. ISTH SSC, May 25–28, Montpellier, France 2016; 14: 52.
49. Pouplard C, Galinat H, Ternisien C, et al. Multicentre evaluation of CK Prest® for assaying plasma levels of factor IX fused with albumin (Idelvion®). *Haemophilia*. 2019;25:e327-e330.
50. Florin L, Devreese KMJ. Comparison of two one-stage clotting assays and one chromogenic FIX assay for monitoring of Idelvion® Replacement therapy. Volume 2, Issue S1. Abstracts of the 64th Annual Meeting of the Scientific Standardization Committee of the International Society on Thrombosis and Haemostasis, July 18–21, 2018.
51. Vangenechten M, Maes M, Maes M, Gadisseur A. Differences in activity of plasma spiked with IDELVION® generated with one-stage clotting assay using different commercial APTT reagents compared with chromogenic method. Volume 2, Issue S1. Abstracts of the 64th Annual Meeting of the Scientific Standardization Committee of the International Society on Thrombosis and Haemostasis, July 18–21, 2018 (Internet). 2018. Available from: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/rth2.12125>.
52. Horn C, Négrier C, Kalina U, Seifert W, Friedman KD. Performance of a recombinant fusion protein linking coagulation factor IX with recombinant albumin in one-stage clotting assays. *J Thromb Haemost*. 2019;17:138-148.
53. Rosén P, Rosén S, Ezban M, Persson E. Overestimation of N-glycoPEGylated factor IX activity in a one-stage factor IX clotting assay owing to silica-mediated premature conversion to activated factor IX. *J Thromb Haemost*. 2016;14:1420-1427.
54. Persson E, La Cour CC. Underestimation of N-glycoPEGylated factor IX one-stage clotting activity owing to contact activator-impaired activation. *Res Pract Thromb Haemost*. 2017;1:259-263.
55. Young G, Ezban M, Clausen WHO, Négrier C, Oldenburg J, Shima M. Chromogenic analysis of FIX activity in haemophilia B patients treated with nonacog beta pegol. *Haemophilia*. 2017;23:e528-e530.
56. Tiefenbacher S, Bohra R, Amiral J, et al. Qualification of a select one-stage activated partial thromboplastin time-based clotting assay and two chromogenic assays for the post-administration monitoring of nonacog beta pegol. *J Thromb Haemost*. 2017;15:1901-1912.
57. Gray E, Kitchen S, Bowyer A, et al. Laboratory measurement of factor replacement therapies in the treatment of congenital haemophilia: a United Kingdom Haemophilia Centre Doctors' Organisation guideline. *Haemophilia*. 2020;26:6-16.
58. Kitchen S, Tiefenbacher S, Gosselin R. Factor activity assays for monitoring extended half-life FVIII and factor IX replacement therapies. *Semin Thromb Hemost*. 2017;43:331-337.
59. Peyvandi F, Oldenburg J, Friedman KD. A critical appraisal of one-stage and chromogenic assays of factor VIII activity. *J Thromb Haemost*. 2016;14:248-261.
60. Hermans C, Altisent C, Batorova A, et al. Replacement therapy for invasive procedures in patients with haemophilia: literature review, European survey and recommendations. *Haemophilia*. 2009;15:639-658.
61. Kitchen S, Signer-Romero K, Key NS. Current laboratory practices in the diagnosis and management of haemophilia: a global assessment. *Haemophilia*. 2015;21:550-557.
62. Marlar RA, Strandberg K, Shima M, Adcock DM. Clinical utility and impact of the use of the chromogenic vs one-stage factor activity assays in haemophilia A and B. *Eur J Haematol*. 2020;104:3-14.

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

How to cite this article: Jeanpierre E, Pouplard C, Lasne D, et al; On behalf of the French Study Group on the Biology of Hemorrhagic Diseases (the BIMHO group). Factor VIII and IX assays for post-infusion monitoring in hemophilia patients: Guidelines from the French BIMHO group (GFHT). *Eur J Haematol*. 2020;105:103-115. <https://doi.org/10.1111/ejh.13423>